



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

Lecture 27: Wave

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Announcement



Scan me!

- Teaching and learning feedback 2
- Completely private and only used to improve the teaching and learning experience!
- Your participation will be highly appreciated.

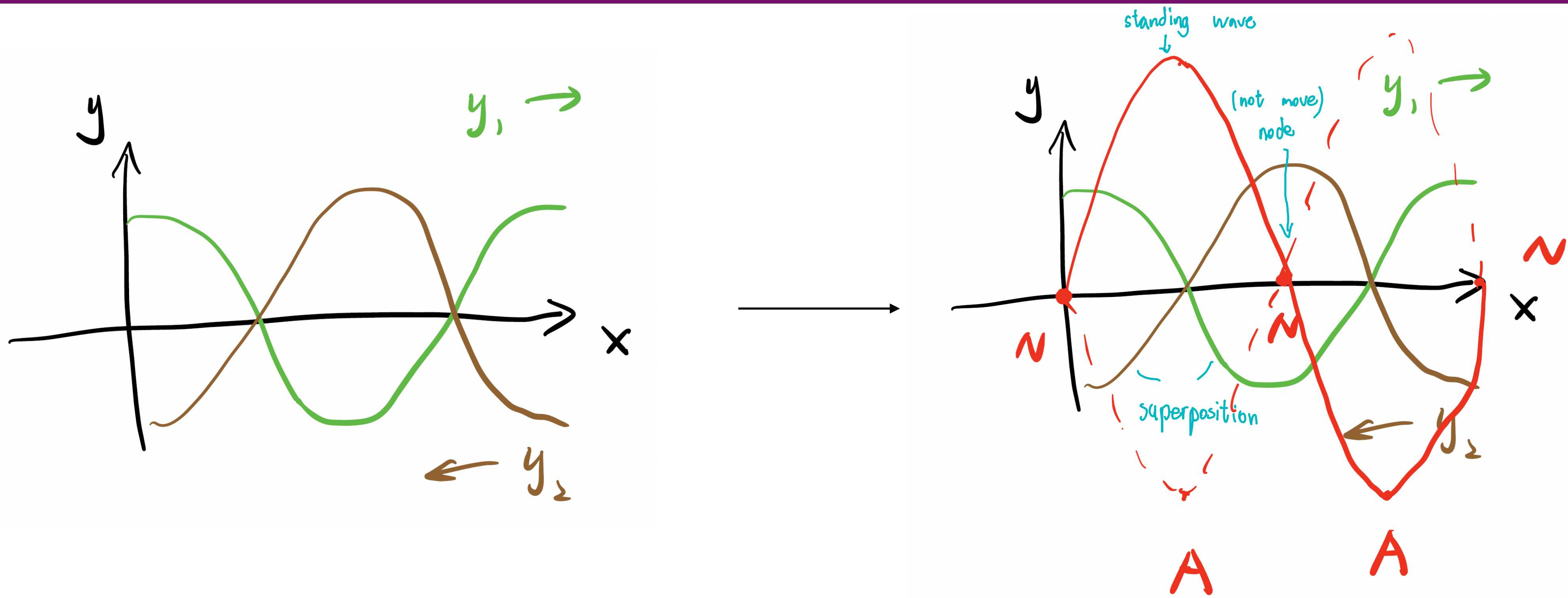
Recap: Transverse Wave

- Wave equation for a mechanical transverse wave: $\frac{\partial^2 y}{\partial t^2} = \frac{F_T}{\mu} \frac{\partial^2 y}{\partial x^2}$, where
 - μ : mass per unit length, kg/m
 - F_T : tension in the string
- Wave speed $v = \sqrt{\frac{F_T}{\mu}}$

Recap: Energy & Power in (Traveling) Wave

- Total energy in one wavelength $E = \pi\sqrt{\mu F_T}\omega A^2$
 - $K = \frac{\pi}{2}\sqrt{\mu F_T}\omega A^2$
 - $U = \frac{\pi}{2}\sqrt{\mu F_T}\omega A^2$
- Average power $P_{\text{av}} = \frac{E}{T} = \frac{1}{2}\sqrt{\mu F_T}\omega^2 A^2$

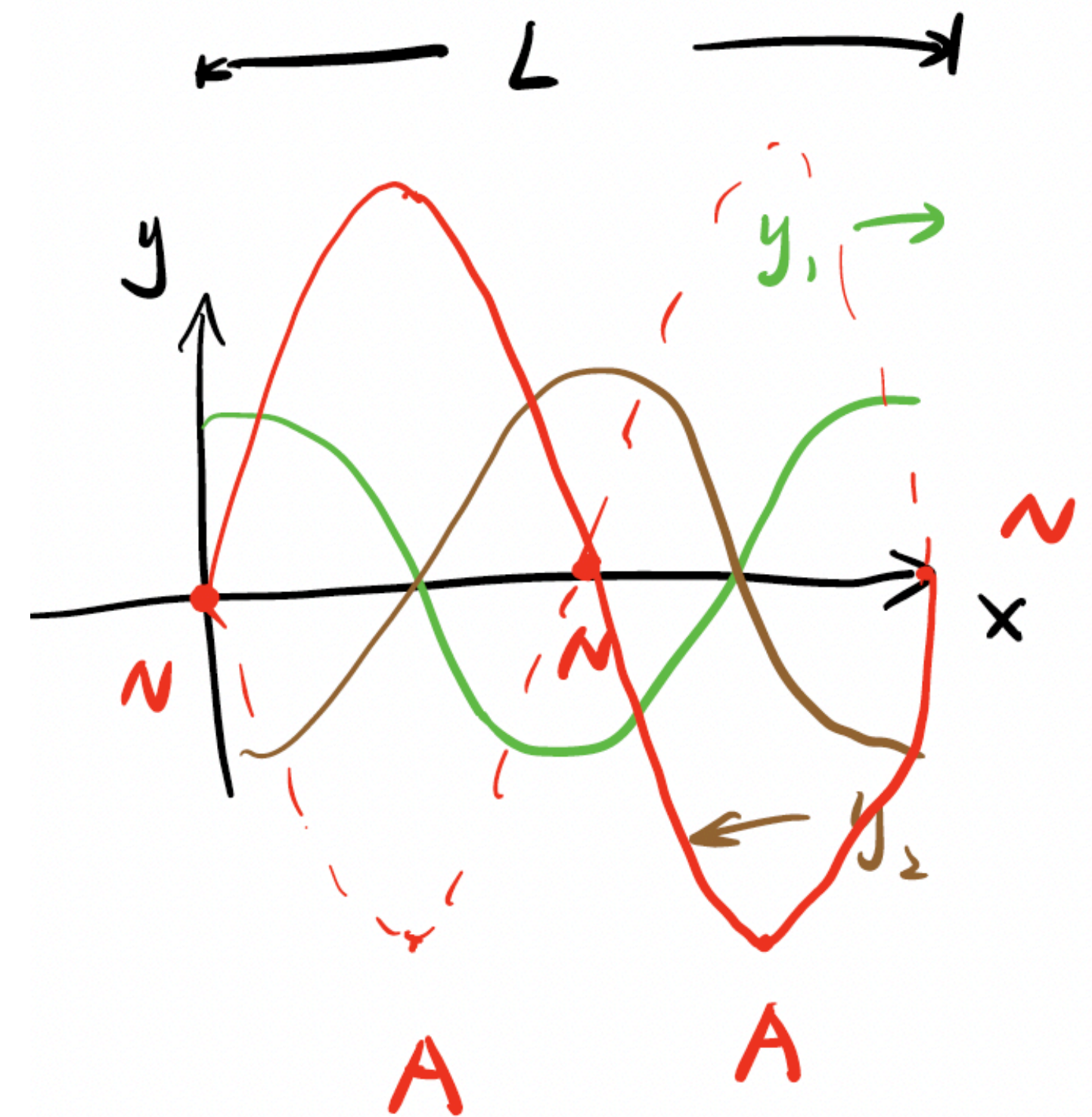
Recap: Standing Wave



- Wave superposition (also called interference): $y(x, t) = y_1(x, t) + y_2(x, t)$ (left-and-right)
- Standing wave: $y = y_1 + y_2 = 2A \sin(kx)\sin(\omega t)$, where (up-and-down only)
 - $y_1 = A \cos(kx - \omega t)$, $y_2 = -A \cos(kx + \omega t)$

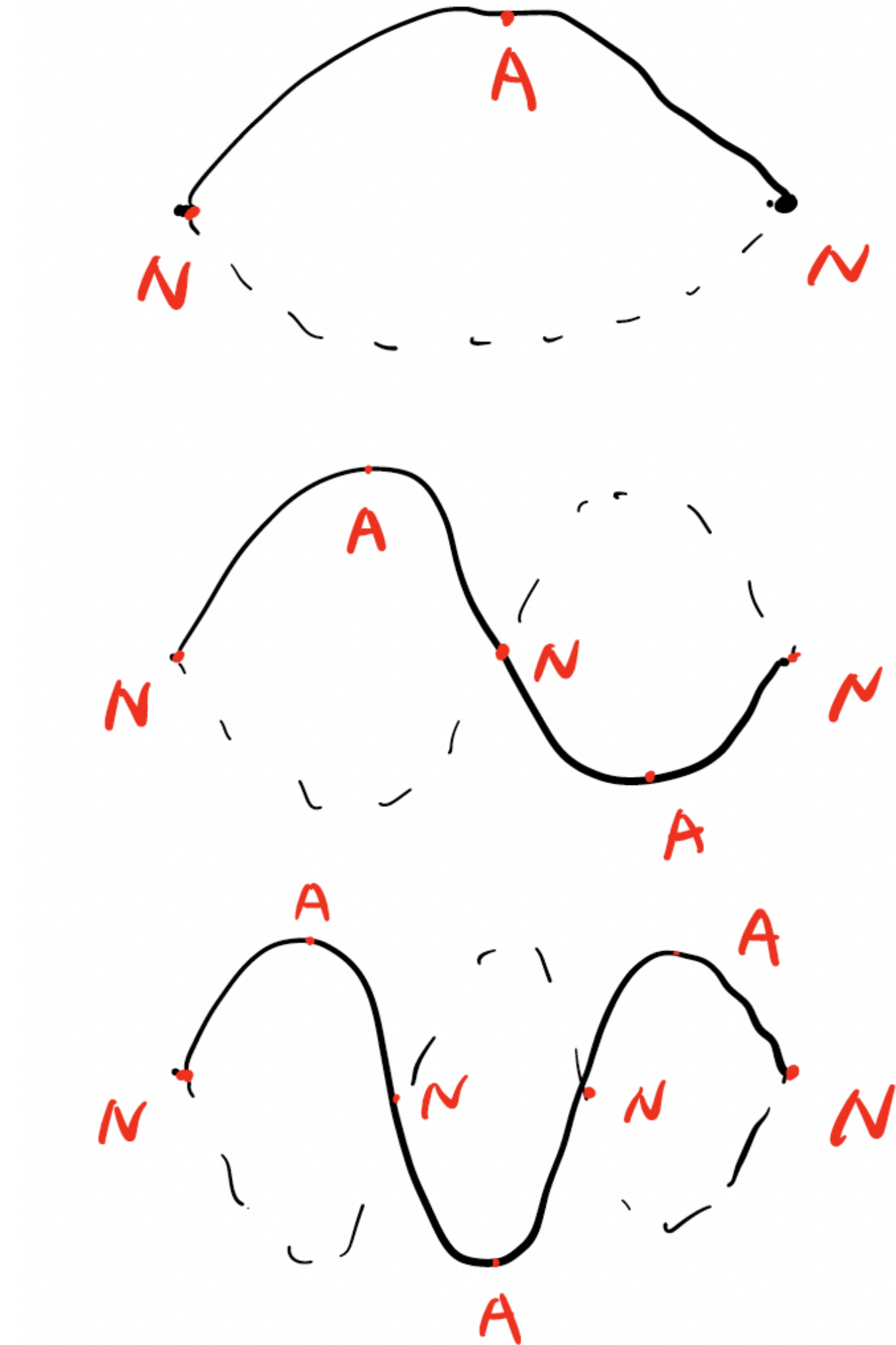
Recap: Standing Wave

- Boundary condition (fixed end at $x = 0$ and $x = L$): $\sin(kL) = 0$, leading to normal modes
 - $\lambda_n = \frac{2L}{n}, n = 1, 2, 3, \dots$
 - $f_n = n \frac{v}{2L} = nf_1, n = 1, 2, 3, \dots$
 - $n = 1$: fundamental mode
- Node (N): destructive interference
- Anti-node: constructive interference



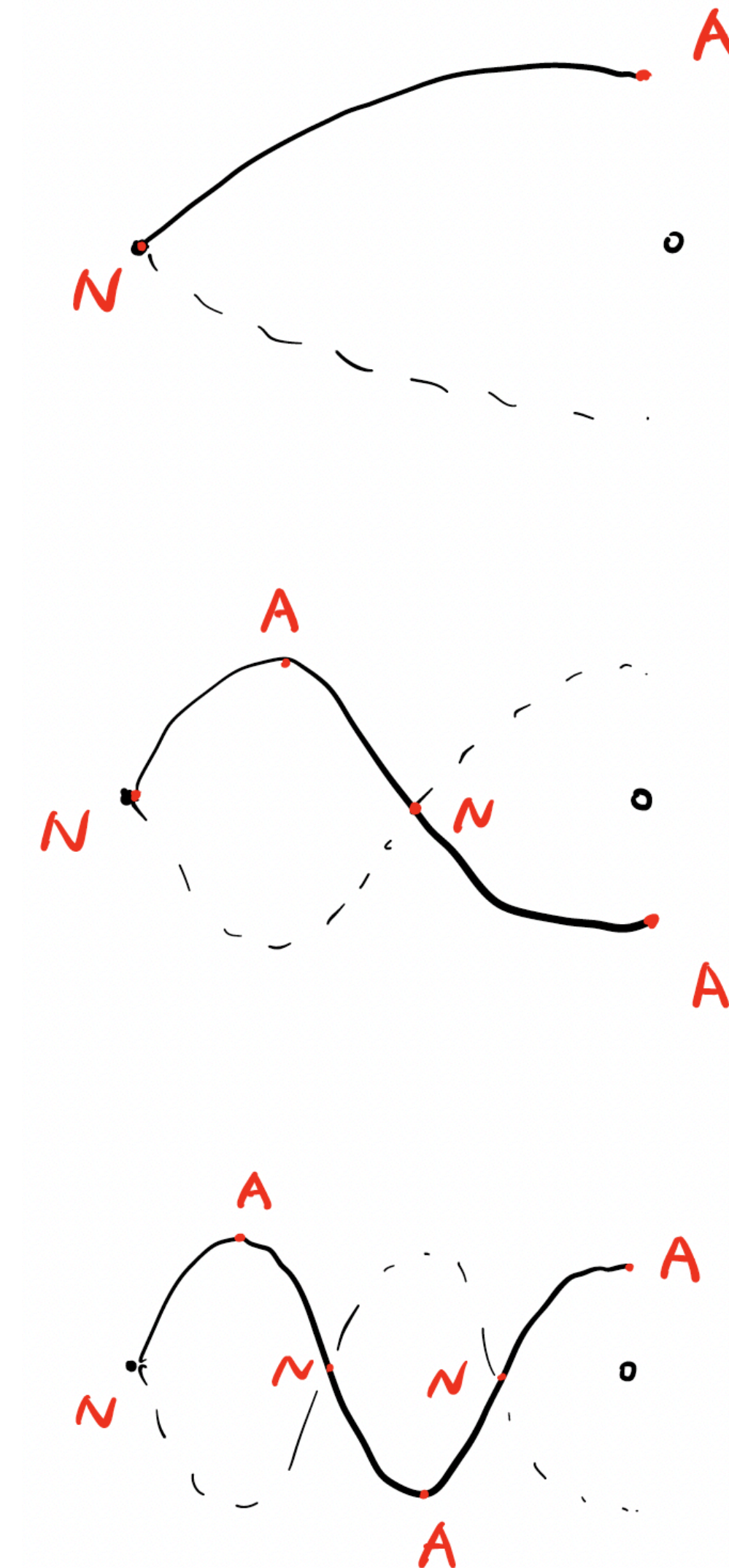
Recap: Standing Wave

- Fixed-Fixed ends
 - Fixed ends: nodes (N)



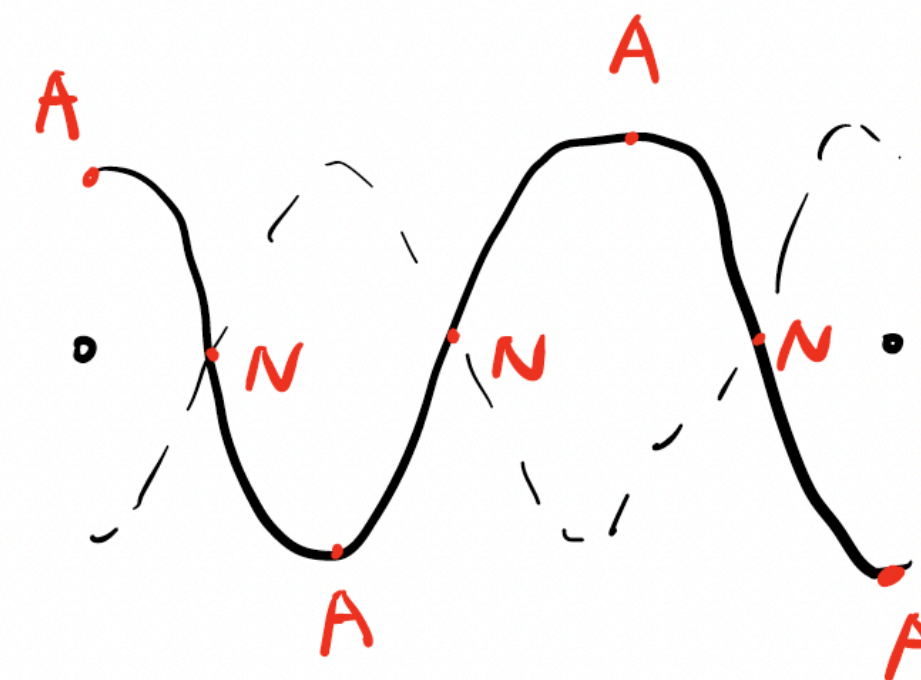
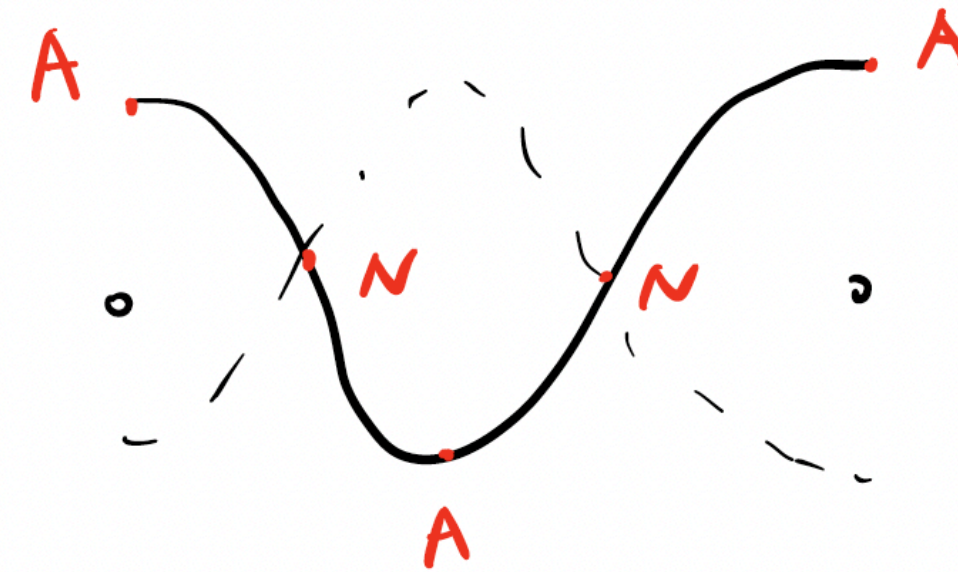
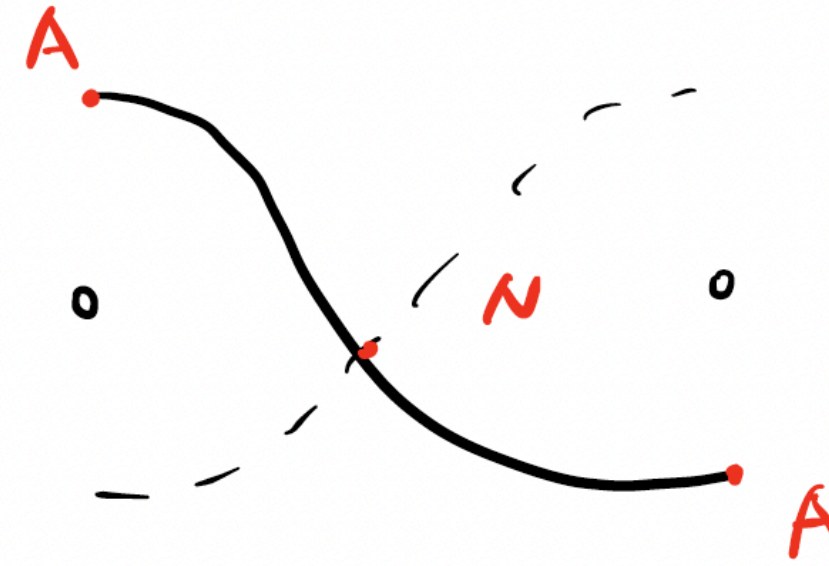
Recap: Standing Wave

- Fixed-open ends
 - Fixed: nodes (N)
 - Open: anti-nodes (A)



Recap: Standing Wave

- Open-open ends
 - Open: anti-nodes (A)



Outline

- Longitudinal wave
- Wave equation and speed of sound wave
- Sound speed
- Power and intensity of sound wave
- Sound standing wave

Longitudinal Wave

- Longitudinal wave: oscillation direction parallel to propagation direction, e.g. sound wave

▸ $y = y(x, t)$, where y and x are parallel.

- $\Delta P = -B \frac{\partial y}{\partial x}$

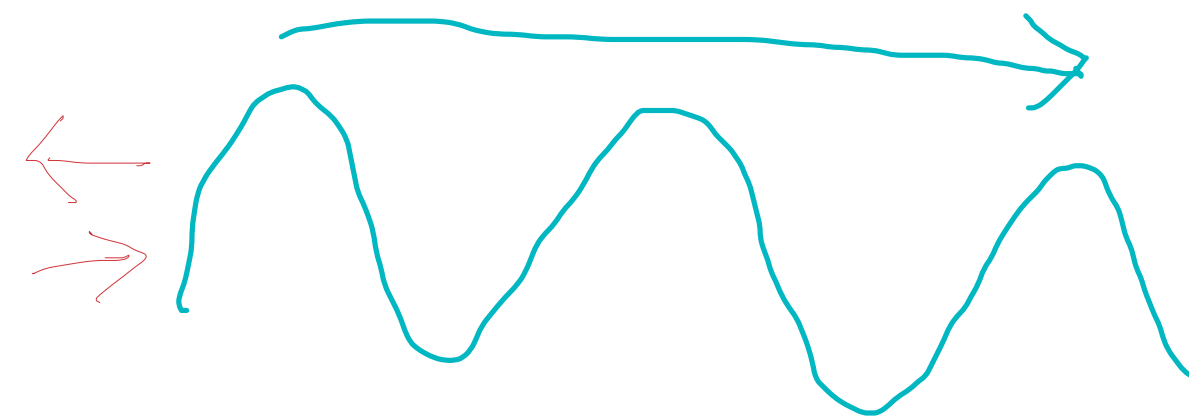
▸ The disturbance is pressure!

$$V = S \cdot \Delta x$$

$$V' = S (\Delta x + y_2 - y_1)$$

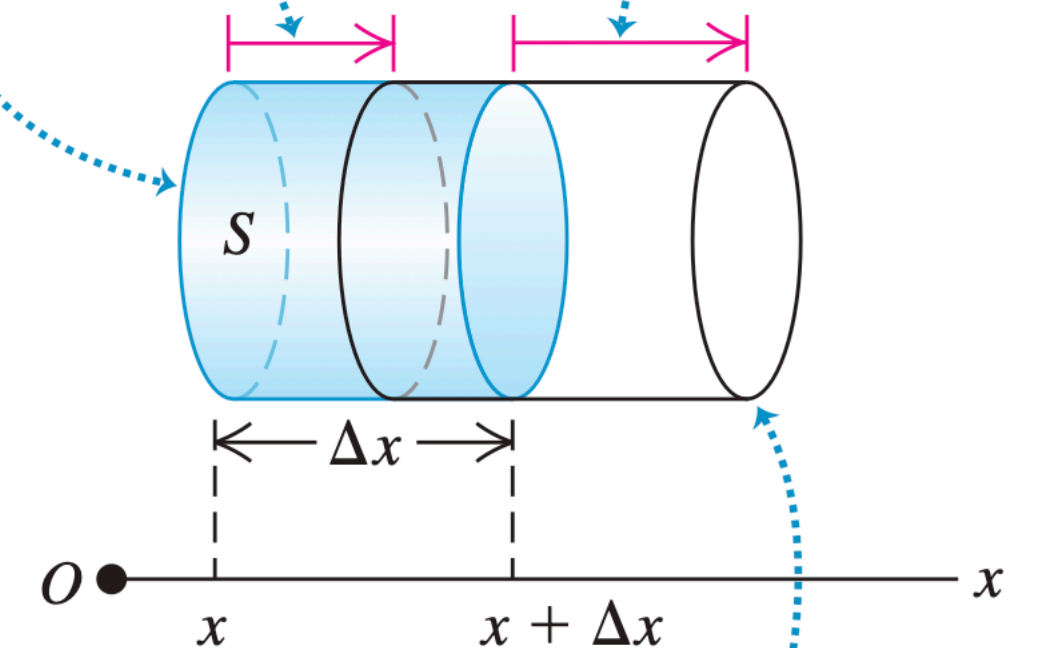
$$\Delta V = V' - V = S \cdot (y_2 - y_1)$$

$$B = - \frac{\Delta P}{\Delta V / V}$$



Undisturbed cylinder of fluid has cross-sectional area S , length Δx , and volume $S\Delta x$.

A sound wave displaces the left end of the cylinder by $y_1 = y(x, t)$... and the right end by $y_2 = y(x + \Delta x, t)$.



The change in volume of the disturbed cylinder of fluid is $S(y_2 - y_1)$.

Wave Equation & Speed

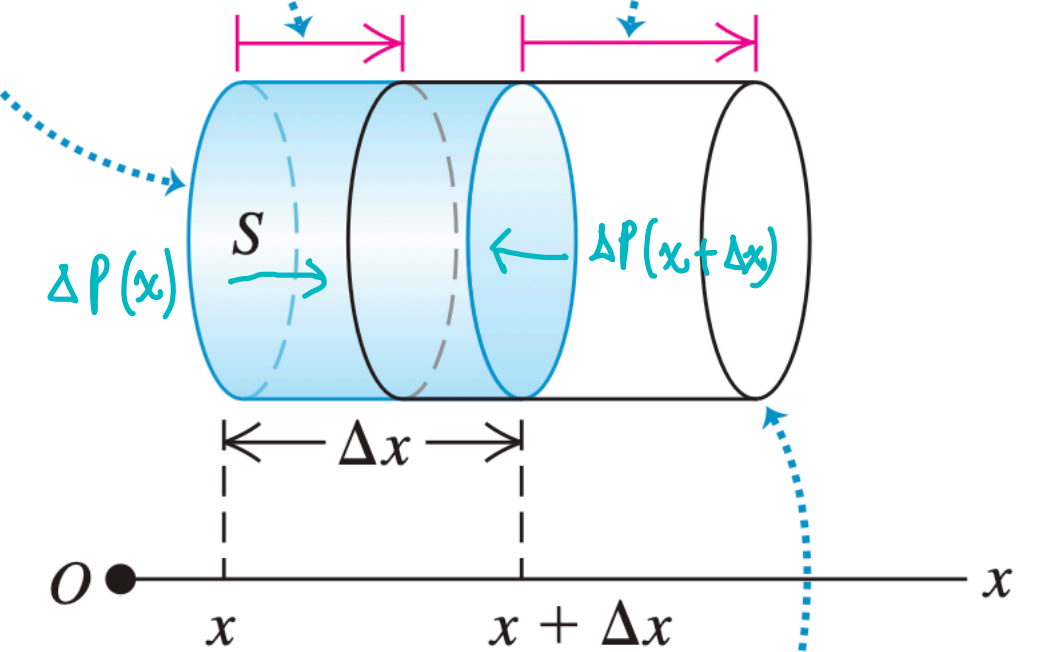
- How to find wave equation and speed for sound wave?
 - Idea: Newton's 2nd law

$$\Delta p = -B \frac{\partial y}{\partial x}$$

$$F = ma$$

Undisturbed cylinder of fluid has cross-sectional area S , length Δx , and volume $S\Delta x$.

A sound wave displaces the left end of the cylinder by $y_1 = y(x, t)$... and the right end by $y_2 = y(x + \Delta x, t)$.



The change in volume of the disturbed cylinder of fluid is $S(y_2 - y_1)$.

Wave Equation & Speed

Sound wave

- Sound wave is a longitudinal wave
- Wave equation: $\frac{\partial^2 y}{\partial x^2} = \frac{\rho}{B} \frac{\partial^2 y}{\partial t^2}$
 - Same solution as before:
 $y(x, t) = A \cos(kx \pm \omega t)$
- Wave (sound) speed:
 - $v = \sqrt{\frac{B}{\rho}}$ (fluid)
 - $v = \sqrt{\frac{Y}{\rho}}$ (solid)

String wave (for comparison)

- String wave is a transverse wave
- Wave equation: $\frac{\partial^2 y}{\partial x^2} = \frac{F_T}{\mu} \frac{\partial^2 y}{\partial t^2}$
- Wave speed: $v = \sqrt{\frac{F_T}{\mu}}$

Sound Speed in the Air

- Sound speed: $v = \sqrt{\frac{B}{\rho}}$. Easy to find ρ , but how about B ?



Isaac Newton
(1643 - 1727)

6. ** Bulk Modulus of an Ideal Gas.

The equation of state (the equation relating pressure, volume, and temperature) for an ideal gas is $pV = nRT$, where n and R are constants.

(a) Show that if the gas is compressed while the temperature T is held constant, the bulk modulus B is equal to the pressure p .

(b) When an ideal gas is compressed without the transfer of any heat into or out of it, the pressure and volume are related by $pV^\gamma = \text{constant}$, where γ is a constant having different values for different gases. Show that, in this case, the bulk modulus is given by $B = \gamma p$.



Pierre-Simon Laplace
(1749 - 1827)

HW8, Q6

Sound Power

v_x is how the energy transport

- Idea:
 - $P = Fv$ v_x v_y

Sound Intensity

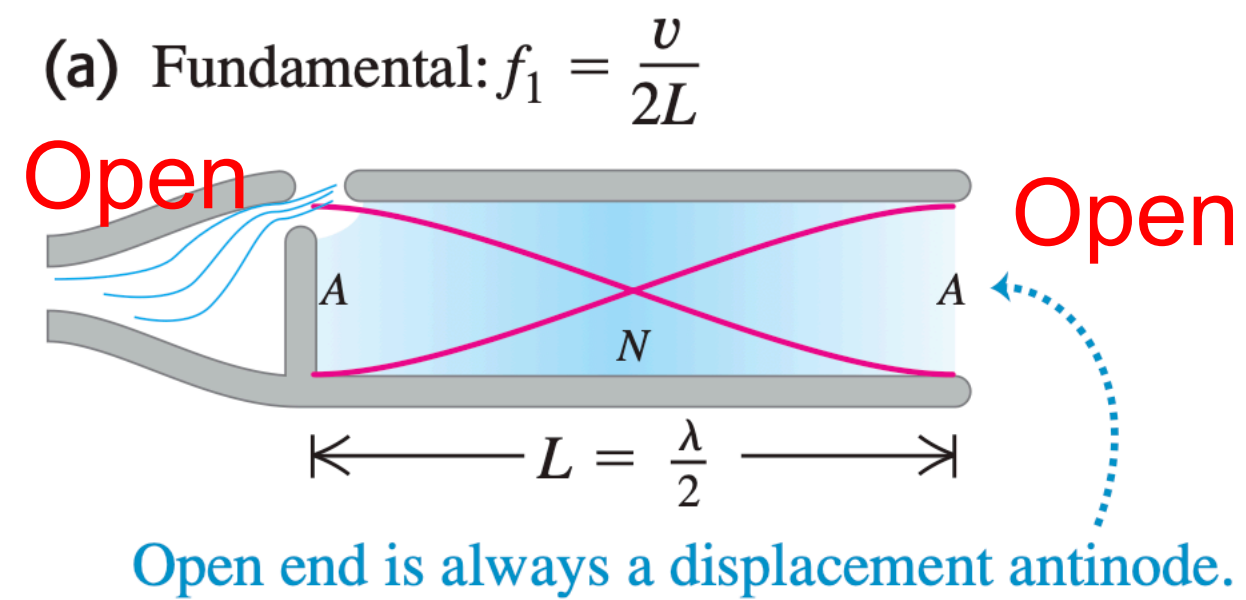
- Intensity (I) = power / area, unit: W/m^2
- Decibel scale $\beta = 10 \text{ dB} \frac{I}{I_0}$, where
 - $I_0 = 10^{-12} \text{ W}/\text{m}^2$

Power and Intensity of Sound

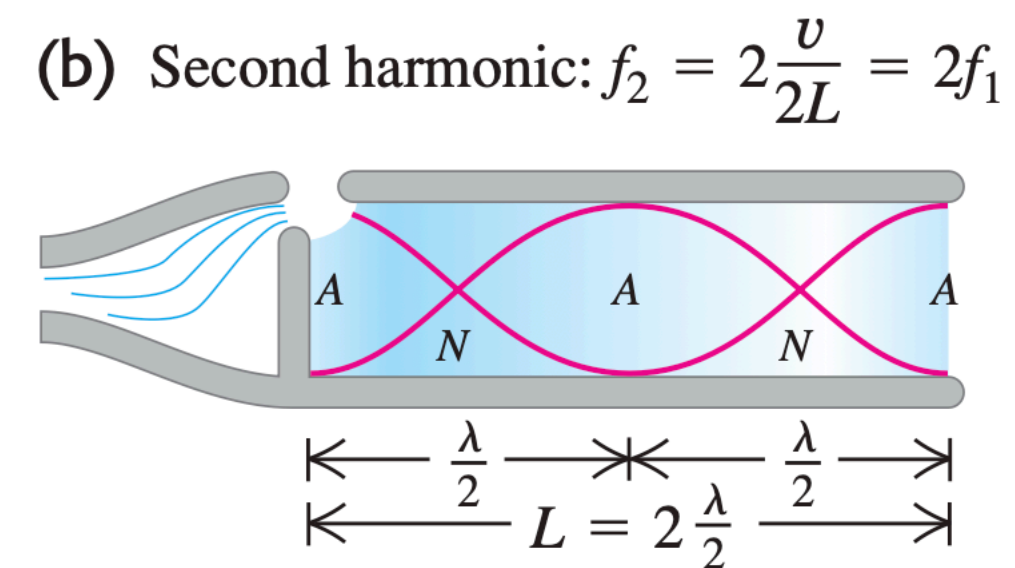
- Sound wave
- Average power $p_{av} = \frac{1}{2}\omega^2 A^2 \sqrt{\rho B} S$, where
 - ω : angular frequency
 - A : amplitude
 - ρ : medium density
 - B : bulk modulus
 - S : area
- Intensity $I = \frac{1}{2}\omega^2 A^2 \sqrt{\rho B}$
- String wave (for comparison)
- $p_{av} = \frac{1}{2}\omega^2 A^2 \sqrt{\mu F_T}$

Standing Wave of Sound

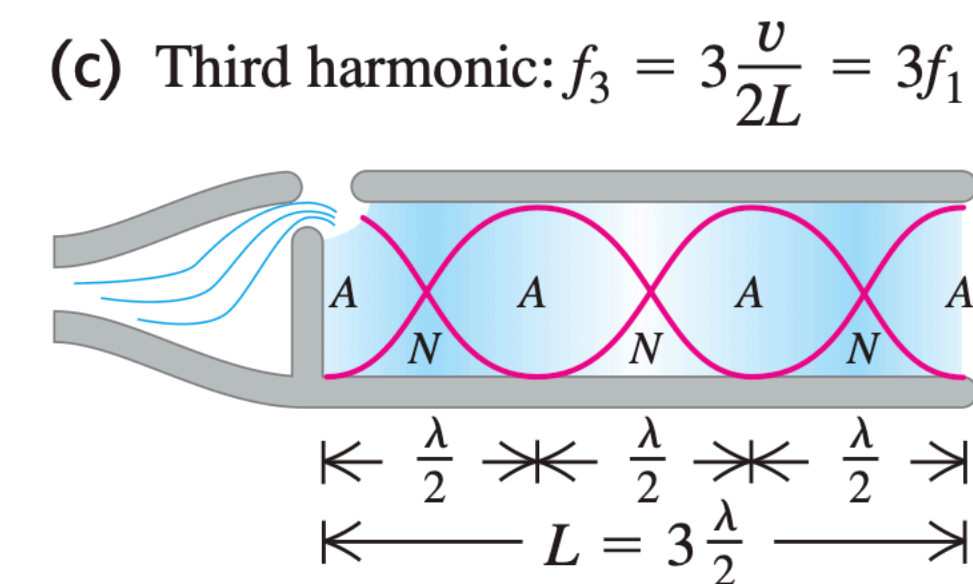
- Open end: anti-node (A)
- Closed end: node (N)



$$\lambda_1 = 2L$$



$$\lambda_2 = L$$



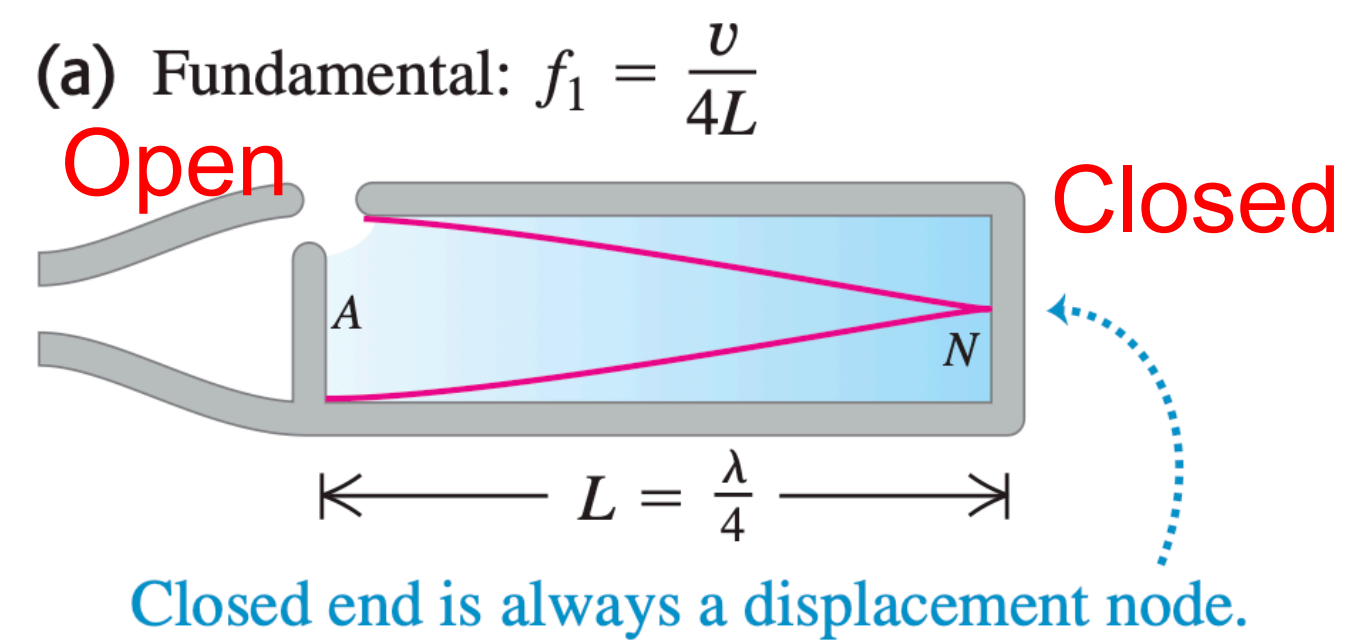
$$\lambda_3 = \frac{2L}{3}$$

For n -th harmonic:

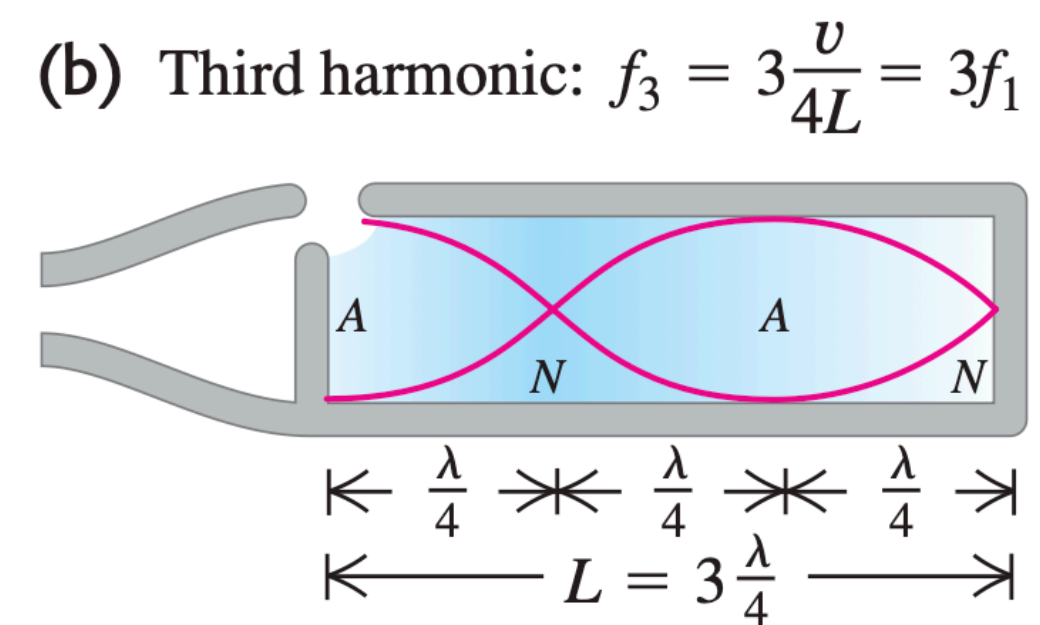
- $\lambda_n = \frac{2L}{n}, n = 1, 2, 3, \dots$
- $f_n = \frac{nv}{2L}, n = 1, 2, 3, \dots$

Standing Wave of Sound

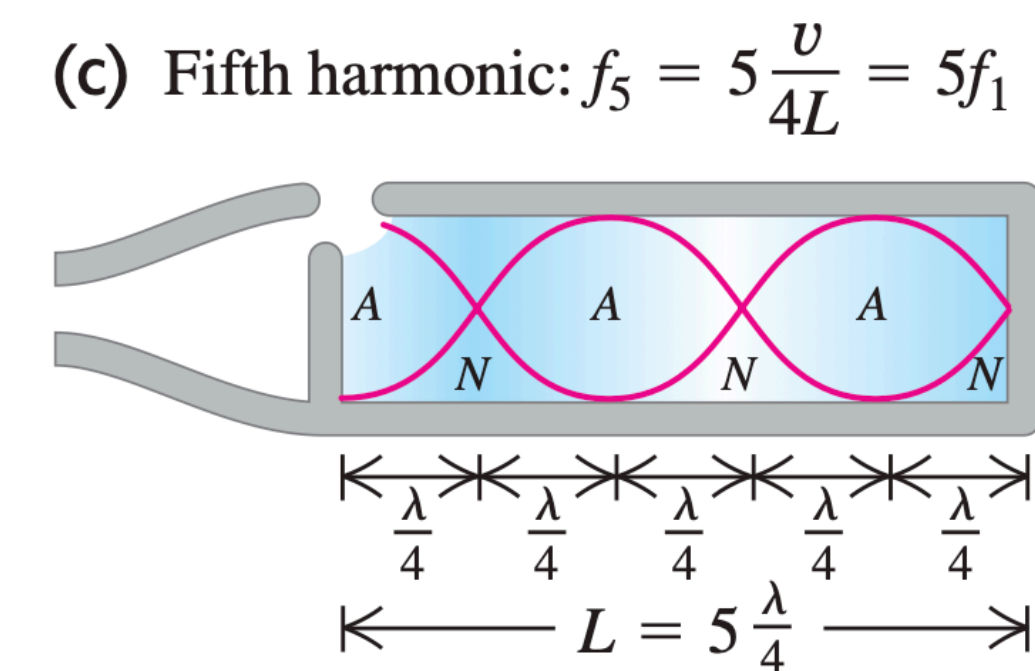
- Open end: anti-node (A)
- Closed end: node (N)



$$\lambda_1 = 4L$$



$$\lambda_2 = \frac{4L}{3}$$



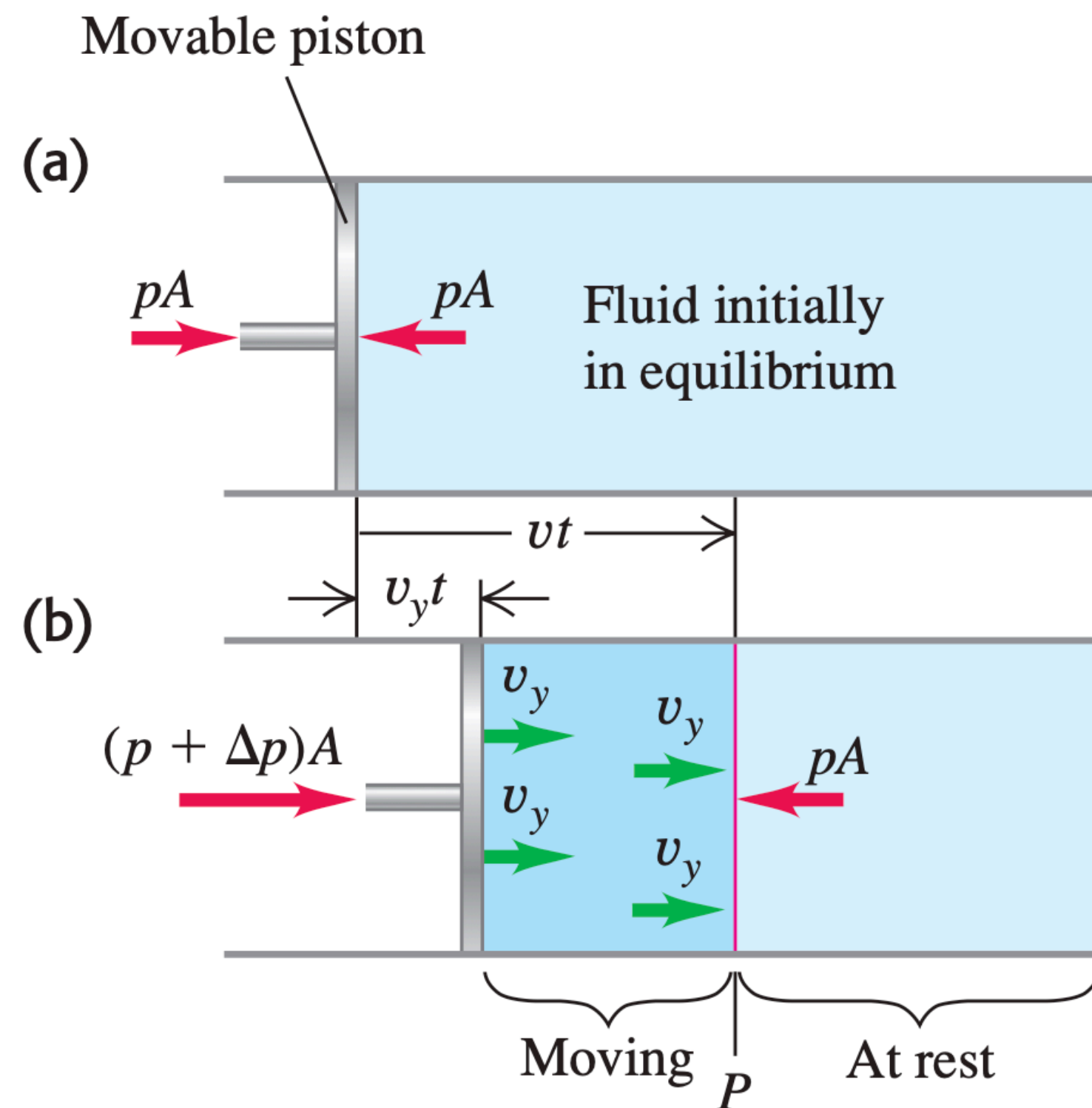
$$\lambda_3 = \frac{4L}{5}$$

For n -th harmonic:

- $\lambda_n = \frac{4L}{n}, n = 1, 3, 5, \dots$
- $f_n = \frac{nv}{4L}, n = 1, 3, 5, \dots$

Back up

Second Method for Wave Speed



- Idea
 - Impulse-momentum theorem
 - Definition of bulk modulus